

Medical Electronics Shaping the Future of Healthcare

Medical electronics has immense potential to address healthcare concerns and challenges like rising costs, access and quality of care



GE Vscan ultrasound machine that fits in the palm

FICCI MEDICAL ELECTRONICS FORUM

Referring to applications of medical electronics for chronic disease management, post-operative care and general wellness, it is well known that portability of medical electronic devices is the key to accessibility and affordability of health diagnostics and other medical solutions. The portability and customisation of medical electronic devices has been made possible by advancements in the

electronics technology.

Key enablers

With newer chip designs and product development due to a huge demand in consumer and mobile electronics, the future is looking bright with designs based on smaller footprints, higher efficiencies and much more functionalities.

Even the design and size of AC-DC and DC-DC converters used in power supply modules is getting reduced due to better component availability and specifications. The innovative selection of ceramics and other materials in place of capacitors and other discrete components leads to smaller sizes and better ratings to accommodate.

The designers' choice is gradually moving from highly flexible field-programmable gate array (FPGA) to application-specific integrated circuit system-on-chip (ASIC-SoC) and application-specific standard product (ASSP), which includes networking, telecommunications, computing, signal processing, analogue-to-digital converters, wireless and medical applications on the same semiconductor with low power consumption.

With a lot to be seen in the times

to come, there is a clear advantage of SoCs whereby dynamic system designing can be highly optimised resulting in cost-effective final medical electronics application with inbuilt functions of device diagnostics and self-test algorithms for error-free service. SoC-based solutions fit aptly in medical electronics applications as these serve a wide range of functionality at low power consumption, robustness, reliability and portability resulting in better return on investment (ROI).

Research and development of advanced algorithms and image processing techniques has resulted in a low-dose, high-contrast computed tomography (CT) solution for the patients' benefit. Clarity of CT scan has improved with introduction of nanopanel detectors and now the scanner can go up to 320 slices.

Especially in medical imaging applications, the flexibility offered by ASIC comes in handy when dedicated core data paths can be customised and designed for processing heavy data flow in the range of 2 GB per second. So application-specific scalability can be achieved for varying data flows in different applications including X-ray, positron emission tomography (PET), CT scan and magnetic resonance imaging (MRI). The performance levels achieved from ASICs are much higher than any other software-programmable chip architecture or fixed-use central processing units (CPUs). However, complex programmable logic devices (CPLDs) and FPGAs are catching up with a benefit of flexibility to change post-design with minor changes in the software.